

# Ben M. Andrew

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## Education

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**University of Manchester**, PhD in Computer Science Sep 2024 – Present

- Research focused on formal verification of degraded systems.
- Supervised by Dr. Marie Farrell, Dr. Louise Dennis, and Prof. Michael Fisher.

**University of Cambridge**, BA (Hons) in Computer Science Oct 2019 – Jun 2022

- Grade II.1 (70%).
- Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing with traditional protocols. Supervised by Dr. Jackson Woodruff.

## Publications

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- Andrew, B. M., Dennis, L. A., Fisher, M., Farrell, M. *Counterexample-Guided Interval Weakening*. Rigorous State-Based Methods (ABZ) 2026. 10.1007/978-3-032-26752-8\_1
- Andrew, B. M. *Weakening Goals in Logical Specifications*. Rigorous State-Based Methods (ABZ) 2025. 10.1007/978-3-031-94533-5\_22

## Skills

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- **Programming Languages:** OCaml, C++, C, Python, Rust
- **Tools:** Git, Docker, Claude Code
- **Spoken Languages:** English, French

## Industry Experience

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**Software Engineer**, Tarides – Paris, France Sep 2022 – Jun 2024

- Built open-source CI infrastructure for the OCaml ecosystem, distributing dependency solving, compilation, and testing across a broad matrix of operating systems, CPU architectures, and compiler versions.
- Ran continuous testing of tens of thousands of packages across the Opam Repository, OCaml's central package universe.
- Exposed concrete bugs in the OCaml 5 multicore runtime through systematic stress-testing, reporting and tracking them to resolution upstream.

**Embedded Systems Intern**, Tarides – Paris, France Jul – Sep 2021

- Research & development project to cross-compile the OCaml bytecode runtime to an ARM-based microcontroller using a real-time operating system as a base layer.

**Computer Graphics Research Intern**, University of Cambridge Jun – Sep 2020

- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

## Teaching Experience

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**University of Cambridge**, Supervisor (TA equiv.) Oct 2022 – Present

- Leading small group supervisions for undergraduate courses in computer science, including topics such as logic, formal verification, and semantics of programming languages.

**University of Manchester**, Graduate Teaching Assistant Oct 2024 – May 2026

- Assisting in teaching graduate and undergraduate courses in computer science, including topics such as software engineering, logic, and formal verification.